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BIOMEDICAL LIMITED

INSTALLATION QUALIFICATION (IQ)

PROTOCOL
FOR
IVEN 4000 LITERS MIXING TANK

EQUIPMENT SR. NO. –

EQUIPMENT ID. –

LOCATION – INFUSION FACTORY MIXING ROOM

PROTOCOL No. – BML/PDF/MR/PT/26/001

PROVIDED BY:

BIOMEDICAL LIMTED

KWARA STATE, NIGERIA

CONTROL COPY

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1.0. PROTOCOL FORMAT APPROVAL:

Signatures in this page indicate that this document and its format have been prepared, checked and approved by the personnel concerned. It shall be signed and dated as shown below.

	WRITTEN BY:	**REVIEWED BY:**
Signature		
Position		
Date		

	APPROVED BY:	AUTHORISED BY:
Signature		
Position		
Date		

2.0 OBJECTIVE

The purpose of the study is to ensure proper installation of the fourteen IVEN **4000 LITERS MIXING TANK** at **THE MIXING ROOM (MR), INFUSION FACTORY, BIOMEDICAL LIMITED.**

This study verifies:

- Qualification of the equipment, installation and to ensure that all background information is performed to assure conformance to **BIOMEDICAL LIMITED** and manufacturer's specifications.
- The equipment meets the current good manufacturing practices requirements and all other regulatory obligations
- No unauthorized or unrecorded modifications have taken place.
- Operation procedures/preventive maintenance procedures have been identified and listed.
- All critical instrumentation have been identified
- All utilities are properly connected
- All safety features are accounted for
- All relevant data obtained have been checked

3.0 SCOPE

- This protocol is applicable in carrying out the installation qualification of the IVEN 4000 LITERS MIXING TANK at the Infusion manufacturing site, Biomedical Limited, Kwara State.
- To be performed after the completion of the design qualification
- To be performed at the time of relocation.

4.0 RESPONSIBILITIES AND QUALIFICATION TEAM

The following departments shall be involved in:

DEPARTMENT	RESPONSIBILITY
Engineering Team	Provision of needed utility and machine installation
Head Of Engineering	Review of the Installation Qualification Document
QA –Manager	Final Documentation and system acceptability
Superintendent Pharmacist/factory Manager	Authorization

The following personnel will be involved in the execution of the Qualification

S/No.	DEPARTMENT	NAME	SIGNATURE
1			
3			
4			
5			

5.0 EQUIPMENT IDENTIFICATION AND DESCRIPTION

IDENTIFICATION

Equipment Name	IVEN 4000 LITERS MIXING TANK
Manufacturer's Name	SHANGAI IVEN PHAMATECH ENGINEERING CO. LTD (CHINA)
Supplier's name	SHANGAI IVEN PHAMATECH ENGINEERING CO. LTD (CHINA)
Location of Installation	MIXING ROOM (INFUSION FACTORY)

DESCRIPTION:

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6.0 PREREQUISITE

The equipment shall confirm the specifications and requirements as specified in the Design Qualification

S/N	ACCEPTANCE CRITERIA
1	User Requirement specification
2	Design Qualification

7.0 ABBREVIATIONS AND DEFINITIONS

Abbreviation	Definition
SOP	Standard Operating Procedure
MOC	Material of Construction
HMI	Human Machine Interface
ENGR	Engineering
IQ	Installation Qualification
QA	Quality Assurance
VAC	Voltage Alternating Current
S. S	Stainless Steel
LTD	Limited
V	Voltage
GMP	Good manufacturing practices
AC	Alternating Current
Amps	Ampere
W	Watt
VDC	Voltage Direct Current
DC	Direct Current
MCB	Miniature Circuit Breaker
KW	Kilowatt

PLC	Programmable Logic Control

8.0 REFERENCES:

The list of reference documents, which are referred for preparation and review of this document, is as follows: -

- S.O.P for Installation, Operation & Performance Qualification of equipment & Instrument: BML-QDG-QA-111
- Machine Manual supplied by SHANGAI IVEN PHAMATECH ENGINEERING CO. LTD
- Design Qualification supplied by SHANGAI IVEN PHAMATECH ENGINEERING CO. LTD

9.0 EQUIPMENT INSTALLATION CHECKLIST:

Pre-Installation Consideration

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	There is no observable physical damage to the tank and all the contact parts are free from scratches & damages.			
2	Sufficient room for servicing, handling & maintenance is provided (around the machine)			
3	Check the floor space. It should not be inclined.			

Utilities

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Power Requirement (415VAC 3-Phase 5 wires (L_1, L_2, L_3, N & E) Frequency: 50Hz)			
2	Adequate illumination at the Proposed Machine room			
3	Proper Earthing System available			

Equipment compliance

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Solution Contact Part MOC: SS316L, Legs, shell and outer part MOC: SS304			
2	Dimension: Height: 3450mm, Inner diameter: 1900mm			
3	Name plate present			

Safety compliance

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	All external surfaces are free of sharp edges.			
2	Safety earthling arrangement is available.			

Electrical diagram check

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Verify the electrical panel wiring diagram			
2	Check whether the position of each component in the electrical is correct			
3	Check whether the wiring of the electrical panel is consistent with electrical diagrams			
4	Check if the electrical diagram is attached to the operation manual			

List of Main Components with their material of construction

S/N	Component	Origin/ Configuration	Qualified / Not Qualified	Checked By	Verified By
1	Temperature Sensor	Brand: Jumo, Prod: 23261300, TN: 48492091, Type: 902810/13 (PT100), Temperature Range: -20-150^{0}c, Power Rating: 3w			
2	Pressure Sensor/Transmitter	Brand: Jumo dTrans p30, Type: 404366, -1-5 bar, 4-20mA, TN:45868979, F-Nr: 002000014502350			
3	Diaphragm Pressure Compound Gauge	Brand: Brighty, Model No: PYTH(PYTHN-98.A0.513.F6), Pressure Range: -1-5bar, Accuracy class: +/- 1.6%, Diaphragm Material:			
4	Integrated Sight Glass	Origin: China, MOC: Glass			
5	Sampling Valve	Manufacturer: Yu'an) MOC: 316L, Size: DN50			
6	Pneumatic Diaphragm Bottom Valve	Manufacturer: Yu'an) MOC: 316L, Size: DN50			
7	Rotating Cleaning Ball	Manufacturer: Yu'an MOC: SS316L, PTFE			
8	Rupture/Bursting Disc	Manufacturer: Shanghai Huali Safety Devices co., limited, Serial Number: 24114000-007, specification Code: YF50-0.38-145/KJ, Burst pressure: 3.8 bar, Material: SS316L, Size: 50mm Diameter, Temperature: 145^{0}c			
9	Electrically Heated Respirator with Insulation and PTFE Filter	Manufacturer: Hangzhou Dali, controller model Number: ZLCP-02, Heating Element: fiberglass-reinforced silicon rubber			

9	Mettler Toledo Weighing Module				
10	Sanitary LED sight glass Light	Manufacturer: MCTUN, Model number: MTX/SD-W1, Origin: China, Power Rating: 3w, MOC: SS304, Operating Voltage: AC/DC 12/24v, Patent Number: ZL201621014			
ELECTRICAL CONTROL BOX					
11	MCB	Manufacturer: Schneider, Model: Ic65N, Current rating: 32A, Configuration: 3-pole			
12	SMPS	Manufacturer: Meanwell, Model: NDR-120-24, Input Voltage: 100-240VAC, input Current: 2.6A Output Voltage: 24V, Output Current: 5A, Frequency: 50/60Hz			
13	Protective Circuit Breaker	Manufacturer: Schneider, Model: GV2ME01C,			
14	Contactor	Manufacturer: Schneider, Model Number: LC1D09, Current ratings: 9A, Type: 3-Pole			
15	8 pin Relay	Manufacturer: Schneider, Range: RXM, Coil Voltage Rating: 24 VDC, Type: 8-Pin			
16	14 pin Relay	Manufacturer: Schneider, Range: RXM, Coil Voltage Rating: 24 VDC, Type: 14-Pin			
17	Smart Temperature/Pressure display meter	Model: FY5140S, Signal Input: 4-20mA, power supply: 12-24VDC, Pressure Range: -1 to 5bar, temperature range: -20 to 150^{0}c			
18	Mettler Toledo weighing controller	Brand: Mettler Toledo, Model: IND360, Electrical Rating: 24VDC, Current: 0.5A, Operating Temperature: -10 to 40^{0}c			
19	Cooling Fan & Filter Assembly	Origin: China, Voltage: AC220V, POWER: 22W +/-10%, Frequency: 50/60Hz,			

20	Variable Frequency Drive	Manufacturer: ABB, Model: ACS180, Power Range: 0.25 to 22kw			
21	Solenoid Valve	Manufacturer: AirTAC, Type: 5-port, 2Position (5/2 way), Port size: ⅛”PT threaded connection, Coil rating: DC24V, 3W			
22	Air Regulator	Manufacturer: AirTAC, Port size: ⅛”PT threaded connection, Range: 0 to 10 bars			
23	Load break switch/disconnect	Manufacturer: Schneider Electric, Model: TeSys Vario V1C, Current rating: 32A			
24	Rotary Switch	Manufacturer: Schneider Electric, Model: ZB2-BE101C, Contact Type: Normally Open			
25	Emergency Button	Manufacturer: Schneider Electric, Model: ZB2-BE102C, Contact Type: Normally Closed			
26	Pilot light indicator	Manufacturer: Schneider Electric, Part Number: XB2-BVM3LC, Mounting diameter: 22mm, Series: Harmony XB2, Voltage Rating: 220-230VAC			
27	Power Socket	Manufacturer: Schneider Electric, Model: EA9XN5210, Current Rating: 10A			

10.0 INSTALLATION QUALIFICATION REPORT:

Attachment of all reports written as:

#Attachment 1

#Attachement 2 etc.

Date of Installation:

Date of Completion:

11.0 DEVIATION REPORT

Prepare a Deviation report including the justification of acceptance and impact on the function of the system/equipment

DEVAITIONS:
JUSTIFICATION FOR ACCEPTANCE:
IMPACT ON OPERATION:

Written By: **Date:**

Reviewed By: **Date:**

Approved By: **Date:**

12.0 CONCLUSION

13.0 List of Attachment

Attach all completed, signed data record forms

14.0 REPORT APPROVAL

Written By: **Date:**

Reviewed By: **Date:**

Approved By: **Date:**